

*index, m, n*

*x*

|          |       |                                                                                                                                   |                                      |
|----------|-------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <i>a</i> | $::=$ |                                                                                                                                   |                                      |
|          |       | <i>x</i>                                                                                                                          | $b = x$                              |
|          |       | <b>AA</b>                                                                                                                         | $b = \{\}$                           |
|          |       | <b>A</b> <i>a</i>                                                                                                                 | $b = b(a)$                           |
|          |       | <b>C</b> ( <i>a</i> <sub>1</sub> .. <i>a</i> <sub><i>m</i></sub> ) ( <i>a</i> ' <sub>1</sub> .. <i>a</i> ' <sub><i>n</i></sub> )  | $b = b(a_1..a_m) \cup b(a'_1..a'_n)$ |
|          |       | <b>CC</b> ( <i>a</i> <sub>1</sub> .. <i>a</i> <sub><i>m</i></sub> ) ( <i>a</i> ' <sub>1</sub> .. <i>a</i> ' <sub><i>n</i></sub> ) | $b = b(a_1..a_m) \cup b(a'_1..a'_n)$ |

|           |       |                                                                                                                                       |
|-----------|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| <i>av</i> | $::=$ |                                                                                                                                       |
|           |       | <b>A</b> <i>a</i>                                                                                                                     |
|           |       | <b>C</b> ( <i>av</i> <sub>1</sub> .. <i>av</i> <sub><i>m</i></sub> ) ( <i>av</i> ' <sub>1</sub> .. <i>av</i> ' <sub><i>n</i></sub> )  |
|           |       | <b>CC</b> ( <i>av</i> <sub>1</sub> .. <i>av</i> <sub><i>m</i></sub> ) ( <i>av</i> ' <sub>1</sub> .. <i>av</i> ' <sub><i>n</i></sub> ) |

|                |       |                  |
|----------------|-------|------------------|
| <i>formula</i> | $::=$ |                  |
|                |       | <i>judgement</i> |

|                  |       |  |
|------------------|-------|--|
| <i>judgement</i> | $::=$ |  |
|------------------|-------|--|

|                    |       |              |
|--------------------|-------|--------------|
| <i>user_syntax</i> | $::=$ |              |
|                    |       | <i>index</i> |
|                    |       | <i>x</i>     |
|                    |       | <i>a</i>     |
|                    |       | <i>av</i>    |